Demo: Loading Macros via `include-in-header`

This page demonstrates the `include-in-header` strategy for loading macros. The YAML front matter of this file includes:

```yaml
format:
  html:
    include-in-header:
      - macros-header.html   # MathJax macro config script
  pdf:
    include-in-header:
      - macros.qmd           # raw LaTeX goes directly into preamble
  revealjs:
    output-file: demo-include-in-header-slides.html
    include-in-header:
      - macros-header.html   # loader that reads macros.qmd
  docx: default
```

> **Note (HTML/RevealJS):** `macros-header.html` is a small loader script (not a second macro-definition file). It reads `macros.qmd`, converts macro definitions to a MathJax config object, and sets `MathJax.tex.macros` before typesetting. View the RevealJS slides output.
>
> **Note (PDF):** For PDF, `macros.qmd` can be included directly since its raw LaTeX goes into the document preamble.
>
> **Note:** This demo uses `macros-header.html` / `macros.qmd` because both files are at the root of this repository. When using this repository as a Git submodule (cloned into a `macros/` subdirectory), replace with `macros/macros-header.html` and `macros/macros.qmd`.

The same configuration can be placed in a shared `_quarto.yml` to apply project-wide:

```yaml
format:
  html:
    include-in-header:
      - macros/macros-header.html
  pdf:
    include-in-header:
      - macros/macros.qmd
  revealjs:
    include-in-header:
      - macros/macros-header.html
```

This inserts the macro definitions for each output format:

- **PDF**: `macros.qmd` goes into the LaTeX preamble via `include-in-header` — all macros work, including `\providecommand`.
- **HTML**: `macros-header.html` is inserted via `include-in-header` and loads macros from `macros.qmd` into `MathJax.tex.macros`.
- **RevealJS**: `macros-header.html` does the same loader step (including MathJax v2 compatibility), so slide math uses the same `macros.qmd` source.
- **docx**: Word documents do not use a LaTeX preamble or MathJax header, so `include-in-header` has no effect on math rendering for docx. The `docx: default` entry above produces docx output without macro support.

Example math

The table below uses macros defined with `\def` in `macros.qmd`, which work in all output formats that support MathJax or LaTeX:

Description	Code	Output
Greek: alpha	`$\a$`	α
Greek: beta	`$\b$`	_
Greek: gamma	`$\g$`	γ
Greek: lambda	`$\l$`	
Greek: sigma	`$\s$`	σ
Log-likelihood	`$\llik$`	ℓ
Independent	`$X \ind Y$`	$X \perp\!\!\!\perp Y$
IID distributed	`$X_i \siid \Normal(\m, \ss)$`	$X_i \;\sim_{\text{iid}}\; \mathrm{N}(\mu,)$
Vector: mu	`$\vm$`	$\vec{\mu}$
Hat beta	`$\hb$`	$\hat{\beta}$